

UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

Department / Program: Computer Science

Human Computer Interaction

Research Paper Project Guidelines and Submission Rules

Course Instructor: Mam Iqra Khalil

Submission Deadline: _____
Section: V16

This project requires students to write an original research paper on a topic related to Human Computer Interaction. The final report must be written in LaTeX and must include proper references and in-text citations.

1. Project Overview

Students are required to submit a research paper on a Human Computer Interaction (HCI) topic. The project should demonstrate understanding of HCI concepts such as usability, user experience, accessibility, interaction design, interface evaluation, user-centred design, cognitive load, information architecture, and technology adoption.

The paper must not be a general essay. It should follow a proper academic research paper format with a clear problem, research objective, literature review, methodology, analysis, discussion, conclusion, and references.

2. Learning Objectives

- To understand how users interact with digital systems, interfaces, and applications.
- To apply HCI theories and principles to a real-world usability or user experience problem.
- To develop academic research writing skills using proper referencing and citation methods.
- To conduct a small-scale HCI investigation using survey, interview, observation, usability testing, or prototype evaluation.
- To present findings clearly through tables, figures, analysis, and practical design recommendations.

3. Topic Selection Guidelines

Each topic must be directly related to Human Computer Interaction. Students may select a topic from the suggested areas below or propose another HCI-related topic with instructor approval.

Suggested HCI Topic Area	Example Research Paper Topic
Usability Evaluation	Usability evaluation of a university website, LMS, mobile banking app, or food delivery app.
Accessibility	Accessibility issues in educational websites for users with visual, hearing, motor, or cognitive disabilities.
User Experience	Impact of interface layout, navigation, colour, typography, or icons on user satisfaction.
Mobile Interaction	User experience challenges in mobile applications used by students.
AI and HCI	User trust, transparency, and usability in AI-based chatbots or recommendation systems.
Educational Technology	Usability of online learning platforms such as LMS, Google Classroom, Moodle, or Microsoft Teams.
Interface Design	Comparison of dark mode and light mode for readability, comfort, and user preference.
Human Factors	Effect of cognitive load, error messages, feedback, and navigation on task completion.

4. Required Research Paper Structure

1. Title Page: University name, subject, project title, student name, registration number, section, instructor name, and submission date.
2. Abstract: A concise summary of the aim, method, key findings, and conclusion.
3. Keywords: 4 to 6 relevant keywords related to the topic.
4. Introduction: Background, problem statement, research gap, aim, objectives, and research questions.
5. Literature Review: Critical discussion of existing studies, HCI theories, models, and related work.

6. **Methodology:** Explain research design, participants/users, data collection method, tools, procedure, and ethical considerations.
7. **Results and Analysis:** Present findings using tables, charts, screenshots, or usability metrics where applicable.
8. **Discussion:** Explain what the findings mean and link them with HCI principles and previous research.
9. **Design Recommendations:** Provide practical HCI improvements based on findings.
10. **Conclusion:** Summarise the main outcomes, limitations, and future work.
11. **References:** Include all sources cited in the paper using a consistent citation style.
12. **Appendices:** Add survey form, interview questions, consent note, raw responses summary, screenshots, or usability task sheet if needed.

5. LaTeX Formatting Requirements

- The final report must be prepared in LaTeX. Reports prepared only in MS Word will not be accepted unless the instructor gives special permission.
- Students may use Overleaf, TeXstudio, or any other LaTeX editor.
- Use a proper research paper template such as IEEE, ACM, or a clean academic article template, as instructed by the teacher.
- The paper should normally be 6-8 pages, excluding appendices, unless the instructor gives a different page limit.
- Use readable formatting, consistent headings, page numbers, figure captions, table captions, and reference formatting.
- All figures and tables must be numbered and discussed in the text, for example: Figure 1 shows..., Table 1 presents...
- Submit the compiled PDF and LaTeX source files if required by the instructor.

6. Referencing and Citation Rules

- Proper in-text citations must be used throughout the paper.
- Every source listed in the reference list must also appear as an in-text citation in the paper.
- Students should use credible sources such as journal articles, conference papers, books, and official reports.
- Avoid weak sources such as random blogs, unverified websites, and copied online notes.
- Use one consistent referencing style throughout the paper, such as IEEE, ACM, APA, or Harvard, according to instructor preference.
- A minimum of 8-12 academic sources is recommended for a good research paper.

7. Originality, AI, and Plagiarism Policy

Strict Rule: If AI-generated content, plagiarism, copied work, purchased work, or another student's work is found in the submitted project, the student will receive 0 marks for the project.

- The paper must be written in the student's own academic language.
- Copy-paste content from websites, previous reports, AI tools, or classmates is strictly prohibited.
- Paraphrasing without citation is also plagiarism.
- Fake references, fake data, fabricated survey responses, and false screenshots are not allowed.
- Students must be able to explain and defend their paper during viva/presentation if asked.
- Any similarity or AI detection issue may lead to marks deduction, resubmission rejection, or 0 marks according to university/instructor policy.

8. Turnitin Report Requirement

- Students must submit a Turnitin similarity report with the final research paper.
- Students must also submit the AI detection report if required by the instructor.
- The submitted Turnitin report must belong to the same final PDF that is submitted for marking.
- Edited reports after Turnitin checking must be checked again before final submission.
- Reports without Turnitin evidence may be considered incomplete.

9. Required Submission Files

- Final research paper in PDF format compiled from LaTeX.
- LaTeX source file or Overleaf export ZIP, if required.
- Turnitin similarity report and AI/plagiarism checking report.
- Dataset, survey responses, interview notes, usability task sheet, screenshots, or prototype link if used.
- Presentation slides, if presentation/viva is required.

10. Evaluation Criteria

Assessment Component	Description	Marks
Topic relevance	Topic is clearly related to HCI and has a focused research problem.	10
Introduction and objectives	Clear background, problem statement, aim, objectives, and research questions.	10
Literature review	Uses credible sources and critically explains previous HCI work.	15
Methodology	Explains participants, tools, data collection, procedure, and ethics.	15
Results and analysis	Presents useful findings with clear explanation, tables, charts, or usability evidence.	15
Discussion and recommendations	Connects findings with HCI concepts and gives practical improvements.	15
LaTeX formatting and presentation	Uses proper academic formatting, captions, structure, and professional layout.	10
Referencing and originality	Uses correct citations and passes AI/plagiarism checks.	10

11. General Instructions for Students

- Choose a topic early and confirm that it is related to Human Computer Interaction.
- Do not write only a descriptive essay. Include a small user study, usability test, interface analysis, prototype evaluation, or survey wherever possible.
- Keep the scope realistic. A small but well-analysed study is better than a broad and weak topic.
- Use ethical data collection. Do not collect sensitive personal information unnecessarily.
- Mention participant count, age range/category, task details, and analysis method clearly.
- Proofread the final report before submission. Grammar, formatting, citations, and figure/table numbering must be checked.

- Late submission, incomplete files, missing Turnitin report, or wrong format may result in marks deduction according to instructor policy.

12. Final Note

This project is designed to help students connect HCI theory with practical user problems. A good submission should show originality, clear research thinking, correct academic writing, proper LaTeX formatting, and honest evidence of the student's own work.